

CM0859 – MT5009 Type Theory

Appendix: Testing with QuickCheck

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Testing with QuickCheck

A paper

Koen Claessen and John Hughes (2000). QuickCheck: A Lightweight Tool for Random Testing of Haskell Programs. ICFP'00. DOI: <https://doi.org/10.1145/357766.351266>.

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Most Influential ICFP Paper Award 2010[†]

“The techniques described in the paper have spawned a significant body of follow-on work in test case generation. They have also been adapted to other languages. . . ”

[†]See www.sigplan.org/Awards/ICFP/.

Testing with QuickCheck

An open source library

QuickCheck on Hackage.[†]

[†]<http://hackage.haskell.org/package/QuickCheck>.

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Commercialisation

QuviQ (www.quviq.com/).

[†]<http://hackage.haskell.org/package/QuickCheck>.

Testing with QuickCheck

Adaptations

QuickCheck has been ported to various languages (Wikipedia 2025-07-28).

C	C#	C++	CHICKEN	CLOJURE
COMMON LISP	CoQ	D	ELM	ELIXIR
ERLANG	F#	FACTOR	Go	Io
JAVA	JAVAScript	JULIA	LOGTALK	LUA
MATHEMATICA	OBJECTIVE-C	OCAML	PERL	PROLOG
PHP	PONY	PYTHON	R	RACKET
RUBY	RUST	SCALA	SCHEME	SMALLTALK
STANDARD ML	SWIFT	Typescript	VB.NET	VHILEY

Testing with QuickCheck

False positive

The program works properly but the test pointed out a fail:

- There is a bug elsewhere.
- There is an error in the specification.

Testing with QuickCheck

False positive

The program works properly but the test pointed out a fail:

- There is a bug elsewhere.
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False negative

There is a bug in the program but the test passed.

Recall Dijkstra's 1969 famous quote:

"Program testing can be used to show the presence of bugs, but never to show their absence!" (Dijkstra 1970)

Testing with QuickCheck

Example

See demo.

References



E. W. Dijkstra (1970). Structured Programming. In: Software Engineering Techniques (NATO Software Engineering Conference 1969). Ed. by J. N. Buxton and B. Randell, pp. 84–88 (cit. on pp. 7, 8).